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“A systemic and comprehensive view of the emerging Urban Air Mobility ecosystem.”

URBAN AIR MOBILITY

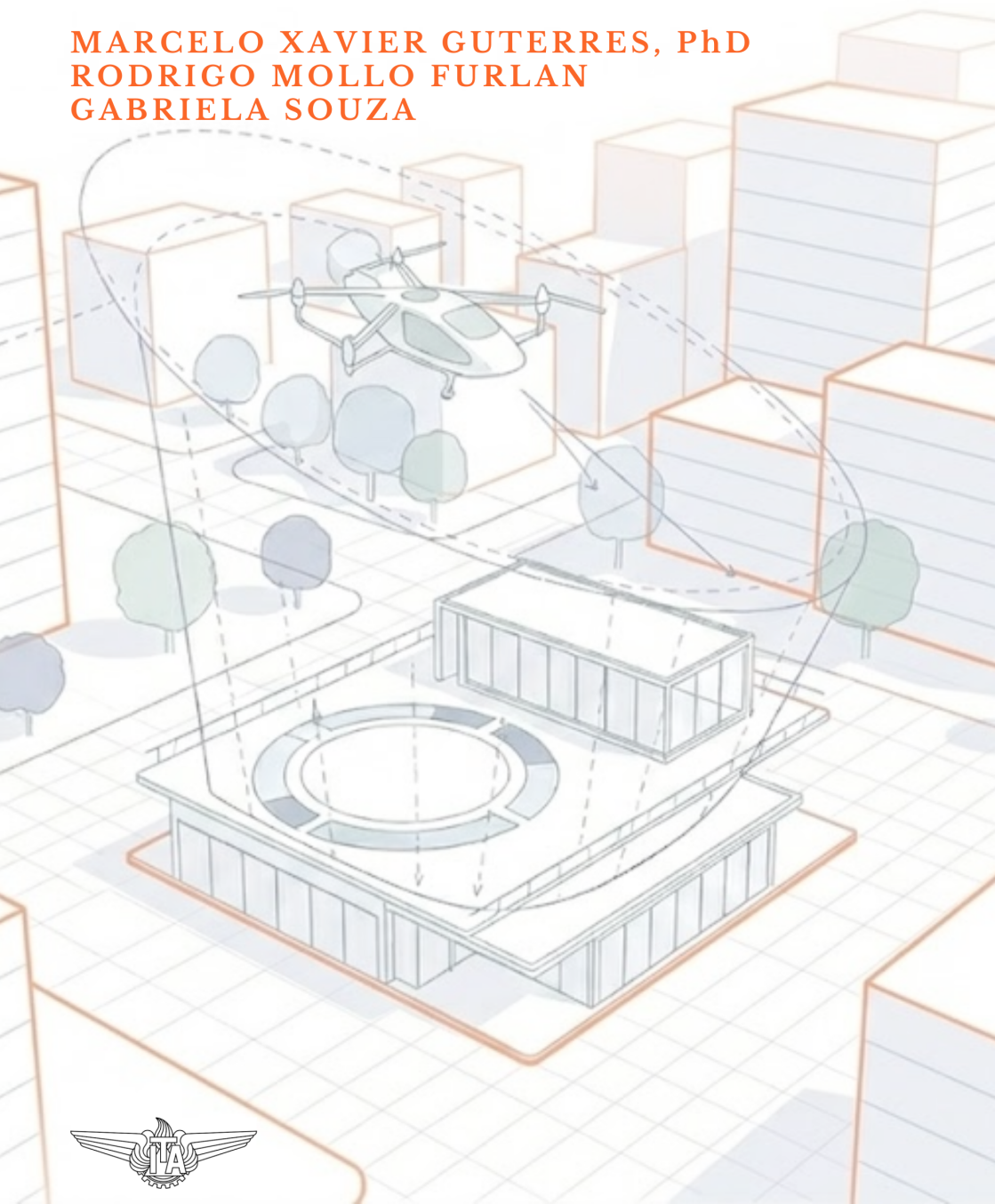


“Modeling, Simulation, and Technical Assessment for the Implementation of Urban Air Mobility.”



URBAN AIR MOBILITY

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"UAM"

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PREAMBLE

Modern life and urban mobility: challenges and impacts

MODERN LIFE imposes several needs on individuals, with **mobility being one of the most frequent**. Whether for work, leisure, or other everyday commitments, these trips often do not occur efficiently, especially in large metropolitan regions. This inefficiency, however, is not limited to densely urbanized areas but also appears in tourist or rural regions, where adequate infrastructure capable of ensuring **efficient mobility** is often lacking.

Solutions to improve travel efficiency are generally not simple, particularly in cities that have grown in a disorganized manner, with historical limitations in territorial planning and modal integration. These **urban phenomena** require structural interventions of great technical and financial complexity, whose effectiveness depends on multiple institutional, economic, and spatial factors.

A classic example of this complexity is the expansion or **construction of new metro lines**. These projects involve high design and construction costs, long implementation cycles, and extended periods for investment recovery. In addition, during their implementation they often generate significant operational impacts, such as partial or total closures of major roadways. Several large-scale metro projects in Brazil illustrate this scenario, highlighting the inherent challenges of expanding infrastructure within consolidated urban environments.

On the other hand, when **transport infrastructure investments** are properly selected and planned, the benefits become evident. The paving of rural roads, for example, can significantly improve access to tourist regions, **reducing average travel time** and **increasing the number of visitors**. Such improvements directly impact the local economy, strengthening regional production chains and expanding territorial connectivity.

Beyond the previous cases, another **relevant challenge associated with mobility** is the insufficiency or absence of structuring infrastructure facilities such as **regional airports**, **adequate bus terminals**, or **passenger port terminals**. This limitation restricts not only everyday mobility but also the economic and tourism development potential of regions.

This reality also produces behavioral and social impacts, including increased stress levels and reduced quality of life associated with prolonged commuting. In summary, **traveling—especially in large urban centers—remains a complex and costly activity across multiple dimensions**, requiring innovative and integrated solutions.

Tangible and intangible costs of mobility

It is equally important to consider the **costs involved in travel, whether tangible or intangible**. Among tangible costs, the average time required to travel between two points stands out, particularly in metropolitan contexts.

With the widespread use of digital systems and navigation applications, an abundance of data on urban travel has become available. Through appropriate statistical techniques, it is possible to estimate the **free-flow travel time** of a route, understood as the travel time observed under operational conditions close to free-flow traffic.

One way to measure the efficiency of urban travel is through the indicator **Additional Travel Time** (T_{AD}), defined by Equation 1:

$$T_{AD} = T_{real} - P_5(T_{trajectory}) \quad (1)$$

Where: T_{AD} represents the Additional Travel Time (minutes); T_{real} corresponds to the actual time spent on the trip under observed condi-

tions (minutes); and $P_5(T_{trajectory})$ indicates the time corresponding to the 5th percentile of the historical set of recorded times for the same route (minutes).

The use of the 5th percentile as an approximation of free-flow time helps reduce the influence of extreme values while preserving a statistically robust estimate of the best operational conditions empirically observed.

Equation 1 allows the efficiency of urban routes to be measured and monitored objectively, offering a comparative basis between real operating conditions and a reference scenario close to free-flow traffic.

Additionally, Figure 1 illustrates this concept, presenting an example in which T_{AD} is obtained from the difference between the travel time observed under congestion (45 min) and the historical 5th percentile of the same route (20 min), resulting in a T_{AD} of 25 minutes.

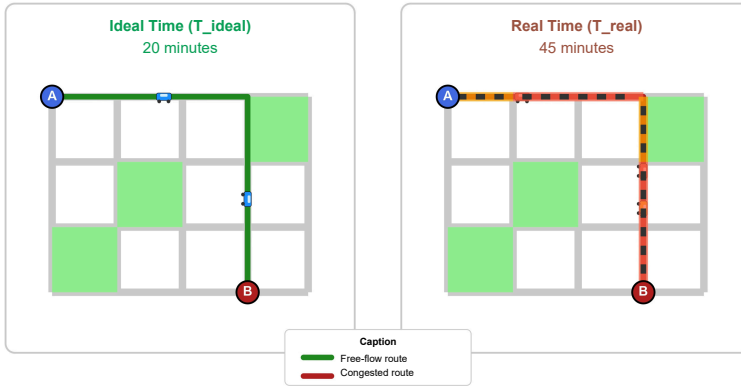


Figure 1: Representation of Additional Travel Time (T_{AD}).

In addition to tangible costs, there are also **intangible costs** that are difficult to measure but equally relevant, such as impacts related to stress, fatigue, and reduced well-being resulting from long daily commuting periods. Studies in urban mobility and public health indicate an association between prolonged travel times and adverse effects on quality of life.

Resistance to the use of public transportation

Another factor influencing the efficiency of urban mobility is the resistance of part of the population to using public transportation. Even when alternatives such as buses, subways, and trains are available, many individuals choose private motorized transportation. Among the factors associated with this choice are perceptions related to comfort, predictability, safety, schedule availability, and control over one's own travel. Behavioral and cultural aspects also play a relevant role in consolidating these modal preferences.

The outlook for ground transportation

Although technological advances and public policies aimed at improving urban mobility exist, structural challenges related to modal integration, infrastructure capacity, and efficient management of travel demand persist. In many metropolitan regions, urban growth has occurred more rapidly than the expansion of transportation infrastructure, generating systems that are often saturated and with significant operational limitations.

In this context, different strategies have been studied and implemented to improve the efficiency of ground transportation. Among them are the expansion and modernization of public transportation systems, the incorporation of digital technologies for traffic management, the development of vehicles with lower emission levels, and the adoption of solutions based on electrification.

However, the adoption of new mobility technologies rarely occurs in a linear manner. Throughout history, several solutions have emerged, been tested, and in some cases temporarily abandoned due to technological limitations or changes in the energy and economic context.

An emblematic example of this phenomenon can be observed in the early twentieth century, a period when electric vehicles coexisted with steam-powered automobiles and the first vehicles equipped with internal combustion engines. At that time, electric automobiles presented some

operational advantages particularly relevant for urban environments, such as silent operation, absence of local emissions, and simpler driving.

Several inventors and engineers dedicated themselves to the development of energy storage technologies aimed at electric mobility. Among them, **Thomas A. Edison** stood out, who in 1911 presented improved versions of nickel–iron rechargeable batteries intended for application in electric vehicles (Figure 2). The objective of these batteries was to increase the durability and reliability of the energy storage systems used in electric automobiles of the time.



Figure 2: Thomas A. Edison next to an experimental prototype battery for electric vehicles, New York, 1911.

Despite the technological advances achieved during that period, factors such as the growing availability of low-cost fossil fuels, the expansion

of refueling infrastructure, and the large-scale production of internal combustion engine vehicles contributed to their dominance throughout the twentieth century.

This historical episode illustrates how the evolution of transportation technologies is strongly associated with the development of adequate energy solutions. In recent decades, significant advances in energy storage systems, power electronics, automation, and computational processing have once again expanded the possibilities for the application of electrification in the transportation sector.

Within this scenario, new mobility architectures are emerging that seek to complement traditional ground transportation systems. Among these initiatives is the development of aircraft capable of vertical take-off and landing, many of them based on electric propulsion, designed to operate in urban and metropolitan environments. These technologies have driven the concept of Urban Air Mobility, discussed in the following sections of this book.

What to expect from Urban Air Mobility

The idea of vertical mobility involving vertical take-offs and landings (VTOL) dates back to the studies of **Leonardo da Vinci**, who around 1490 sketched the concept of the *helix pteron*, often cited as a conceptual precursor to rotary-wing aircraft (Figure 3).

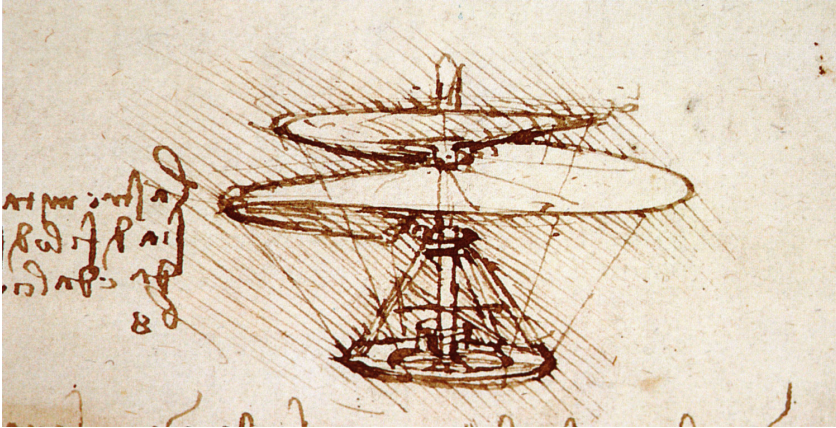


Figure 3: Historical representation of the helix pteron concept.

Only in recent decades, however, have advances in electric propulsion, energy storage systems, flight automation, computational processing, and large-scale data analysis created the technical conditions necessary for the development of operational architectures aimed at using urban airspace for short- and medium-distance travel.

In this scenario, systems with increasing levels of automation, remote supervision, and digital integration are emerging, which may expand access to urban air transportation, provided that the strict certification requirements, operational safety standards, and social acceptability criteria established by competent authorities are met.

What to Expect from This Book

This book presents itself as a **structured technical reference for understanding, planning, and implementing Urban Air Mobility (UAM)**, aimed at industry professionals, researchers, planners, and public agents involved in the formulation of policies and infrastructure projects.

Initially, the **UAM ecosystem** will be discussed, including eVTOL aircraft (*electric Vertical Take-Off and Landing*), operators, infrastructure, air traffic management systems, and relevant institutional stakeholders.

Next, the focus will shift to **landside planning**, addressing demand modeling, integration with existing urban systems, and the incorporation of **vertiports as new traffic-generating hubs** and structuring elements of the territory.

Technical criteria for **vertiport site selection** will be explored, considering aeronautical constraints, regulatory requirements, accessibility, urban integration, and environmental impacts. Aspects related to **vertiport design and engineering** will also be discussed, including civil infrastructure, electrical systems, operational support, and sustainability-oriented solutions.

The book will further advance to **airside planning**, covering airspace organization, operational procedures, and requirements associated with the safety and efficiency of operations.

Each chapter will present applied examples and structured methods, aiming to provide the reader with a consistent technical foundation for the integrated planning of Urban Air Mobility in different territorial contexts.

The central purpose is to provide technically robust and methodologically grounded material, contributing to the critical analysis and responsible implementation of this new mode of transportation.

Ultimately, understanding the technical, operational, and regulatory foundations of Urban Air Mobility is an essential step toward critically assessing the role that this new transport mode may play in the evolution of transportation systems in the coming decades.

Contents

Preamble	i
Contents	x
1 Advanced Air Mobility	1
1.1 AAM Subsystems	2
1.2 Taxonomy of Various AAM Aspects	7
1.3 Regulatory Framework for AAM	8
1.4 Aircraft in the AAM Ecosystem	11
1.5 Integrated Learning Section	14
2 Course Syllabus and Project	18
2.1 Course Description	18
2.2 Activity Schedule	19
2.3 Project	24
Bibliography	28

1

ADVANCED AIR MOBILITY

AAM is an **innovative proposal to revolutionize air transportation**. To this end, it will utilize **electric, autonomous, and quieter aircraft**. This new concept is being developed by NASA, FAA, EASA, and ANAC, in partnership with academia and industry. Its **objective is to enable agile and efficient transportation**. To serve areas traditionally neglected by **conventional aviation**, both **urban** and **rural** [6].

A **practical application, for example, will be rapid delivery of medicines to remote regions**. Another possibility will be the **transportation of passengers between airports and urban centers**. These applications will facilitate connections and extend the reach of **conventional air transportation** [6].

Furthermore, AAM will enable **diverse new types of air operations. Ranging from small local deliveries to regional routes between medium-sized cities**. It will also reach connections between major metropolitan centers.

This model will connect previously inaccessible or underserved locations. It will leverage new technologies to integrate regions. In summary, **it will dynamize urban and rural mobility efficiently**.

Main practical applications of AAM include:

- *Rapid delivery of medicines to isolated regions;*

- *Passenger transportation between urban centers and airports;*
- *Facilitated regional connections between medium-sized cities;*
- *Support in medical emergency situations and rescue operations.*

These applications will **transform the quality of people's lives**. Enabling rapid access to essential resources. Offering critical services with greater efficiency. It will also **reduce travel times and environmental impacts**.

To **operate efficiently**, AAM will utilize existing and new infrastructures, such as: **airports, heliports, and vertiports**. The system will include **new aircraft, use cases, and business models**. Serving **new needs of travelers, consumers, airlines, and infrastructure owners** [3].

From the foregoing, it can be inferred that AAM will present significant opportunities for communities. However, the PAS report [3] identifies several challenges. These include **community impacts, public acceptance, and safety**. Adding concerns regarding social equity, land use, and multimodal integration. All are fundamental factors for effective planning and implementation.

To this end, **engagement of planners and policy makers will be essential**. This engagement will **prepare communities for the impacts of the emerging ecosystem**. This will ensure an **orderly and beneficial transition for all stakeholders**.

Finally, proper planning addresses **multiple cross-cutting issues**. Which include, for example, **mobility, land use, and environmental protection**. Encompassing **financing, economic development, and social equity**. Thus, planners, with their multidisciplinary functions, are strategically positioned. They can and should assist communities in understanding and preparing. They are capable of guiding **AAM growth in a sustainable and inclusive manner** [3].

1.1 AAM Subsystems

AAM will be composed basically of three subsystems. They are:

UAM

Definição. Safe, sustainable, accessible, and economical air transportation in metropolitan areas, focused on passengers, cargo delivery, and emergency services. Operates in urban, suburban, and exurban environments¹

Infraestrutura típica. Vertiports, adapted heliports, landing and take-off points specific for eVTOLs, dedicated areas for drones, and spaces adapted in existing buildings.

Rural Air Mobility

Definição. Air transportation for sparsely populated regions, encompassing passenger mobility, logistics, and medical emergencies. Interconnects rural communities to urban centers and includes specific services, such as agricultural spraying with drones.

Infraestrutura típica. Small aerodromes, heliports, rural airstrips, and areas adapted for emergency landings and takeoffs.

RAM

Definição. Air transportation for medium distances, between 50 and 500 miles (80 to 800 km), connecting intra- and inter-regional areas²

Infraestrutura típica. Regional airports, small airport terminals, prepared runways for small aircraft, and facilities adapted for regional operations.

In addition, H. Pak *et al.* [7] present **four main modalities of urban air transportation services** in Figure 1.1, each focused on **specific mobility needs**.

¹Urban environments refer to densely populated areas with contiguous buildings. Suburban environments are peripheral residential areas with moderate density, and exurban areas are near-rural areas, with low population density, but still under economic urban influence.

²Intra-regional connections refer to links between localities in the same geographic, economic, or administrative region.



Figure 1.1: Main modalities of urban air transportation services.

1. *Sub-Urban-Commuter*

This service will primarily serve people who commute daily from suburbs or peripheral residential areas to the city center. It functions as a **faster alternative to conventional ground transportation**, significantly reducing travel time in congested areas.

2. *Intra-City*

Also known as **air taxi**, this service will operate on-demand between different points within the same city. It offers full flexibility, allowing passengers to **choose any authorized landing location in the urban network**, creating a highly personalized point-to-point transportation network.

3. *Airport-Shuttle*

This service will provide **direct connections between strategic city points and the airport**, typically at predetermined times. Experts predict this modality will likely be the first form of Urban Air Mobility to be implemented at scale, initially focusing on premium airline passengers, and gradually expanding to all travelers.

4. *Inter-City*

This service will connect nearby cities, covering distances greater than strictly urban services. Because it operates on longer routes, it may use both **eVTOL aircraft** and **short takeoff and landing aircraft (STOL)**. This modality approaches the concept of **RAM**, although generally employing smaller and more agile aircraft.

These **four modalities form the foundation of the AAM ecosystem**, offering faster, more efficient, and potentially more sustainable alternatives to conventional transportation means, especially in metropolitan areas with high population density and frequent congestion.

Table 1.1 synthesizes the main use cases, presenting essential parameters, such as operational range and speed. In addition, it includes the ***Metroplex*** variant, defined as an expansion of the *Intra-City* concept applicable to metropolitan regions of large territorial extension, high population density, and elevated operational complexity.

Use Case	Range	Speed	Description
Intra-City	Up to 50 km	Up to 100 km/h	On-demand flights within central and urbanized areas of cities. High traffic density and short-distance operations. Flight mission with up to two intermediate stops without the need for recharging.
Expanded Metroplex	Up to 100 km	Up to 150 km/h	On-demand flights in central areas of large metropolises (megacities). Operations at longer distances than in strictly urban context. Flight mission with none or one intermediate stop without the need for recharging.
Airport Transportation (<i>Airport Shuttle</i>)	Up to 30 km	Up to 150 km/h	Scheduled flights between airports and selected locations (<i>Central Business District- CBD</i>). Vehicles with greater cargo capacity and baggage space. Flight mission between two vertiports with recharging at each segment.
Suburban Commuting (<i>Suburban-Commuter</i>)	Up to 70 km	Up to 150 km/h	Scheduled flights between suburbs or satellite cities and the urban center. Demand concentrated during peak hours and low during intermediate periods. Flight mission between two vertiports with recharging at each segment.
Inter-City	More than 100 km	More than 100 km/h	Scheduled flights between two or more nearby cities. Vehicles for medium/long-distance flights with greater passenger comfort. Flight mission between two vertiports, with recharging after each flight.

Table 1.1: Use cases and technical parameters for Urban Air Mobility.

1.2 Taxonomy of Various AAM Aspects

AAM encompasses a wide range of technologies, propulsion configurations, and applications. Figure 1.2 presents a taxonomy of the main aspects of this ecosystem, organizing them into five main categories: **energy model**, **control model**, **propulsion configuration**, **intelligent systems**, and **applications** [1].

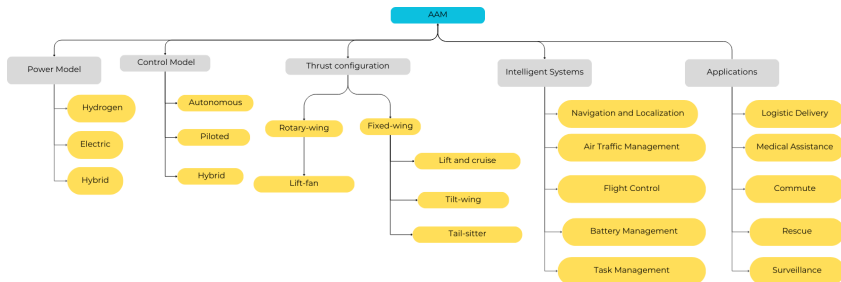


Figure 1.2: Taxonomy of various AAM aspects

The **energy model** defines the propulsion source of aircraft, which may be **hydrogen, electric, or hybrid**, reflecting trends in sustainable aviation and the pursuit of greater energy efficiency. The **control model** classifies aircraft as autonomous, piloted, or hybrid, highlighting technological evolution in flight management and the integration of artificial intelligence for aircraft control.

The **propulsion configuration** differentiates aircraft with rotary wings and fixed wings, **directly influencing their aerodynamics and operation**. Among fixed-wing aircraft, there are subdivisions such as *Lift and Cruise*, which separates takeoff and cruise systems, *Tilt-wing*, which allows variation in wing inclination, and *Tail-sitter*, which enables vertical landing and takeoff positioning.

Intelligent systems encompass technologies for safe and efficient operation of AAM aircraft, including **navigation and localization, air traffic management, flight control, battery management, and task management**. These innovations ensure the viability of autonomous operations and **integration with existing air control systems**.

Finally, **AAM applications** span diverse areas, such as **logistics delivery, medical assistance, urban transportation (commute), rescue, and surveillance**. These applications evidence the positive impact of AAM on urban mobility and essential services, promoting greater accessibility and efficiency in air transportation.

In summary, the taxonomy presented in Figure 1.2 provides an overview of AAM's fundamental elements, assisting in understanding the challenges and opportunities of this emerging sector.

1.3 Regulatory Framework for AAM

To date, there is no consolidated regulatory framework for AAM. However, some progress has been made, and countries that have begun developing policies to govern this ecosystem have concentrated their efforts on three fundamental pillars:

- **Aircraft and airworthiness requirements;**
- **Airspace management;** and
- **Operational requirements.**

A typical AAM ecosystem is illustrated by Figures 1.3 through 1.5, structured from existing regulations, especially those applicable to helicopters and fixed-wing aircraft.

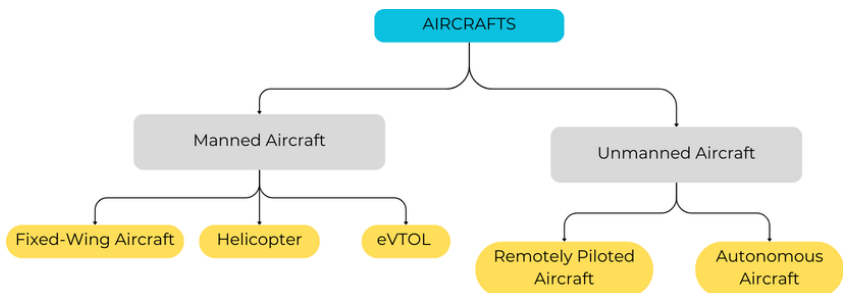


Figure 1.3: Classification of Aircraft in the AAM Ecosystem.

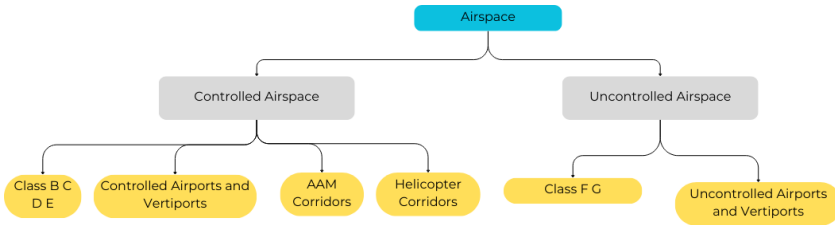


Figure 1.4: Classification of Airspace in AAM.

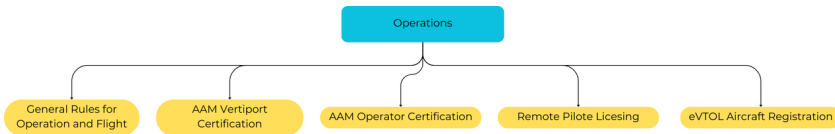


Figure 1.5: Operational Rules and Certifications in AAM.

Although AAM promises a revolution in urban transportation, bringing benefits such as **enhanced connectivity and potentially more sustainable operations**, there are critical challenges to overcome before this technology becomes a consolidated reality. Among them, the following stand out:

- **technical regulations that ensure operational safety**; and
- **airspace management**, since current ATM systems were not designed to integrate aircraft operating at lower altitudes, where drones and eVTOLs will play a central role.

To establish a **regulatory framework**, covering aspects such as **aircraft certification, ground infrastructure, and airspace integration**, a market study conducted by McKinsey [4] highlights five categories (Figure 1.6) of fundamental regulations needed to enable services such as **last-mile delivery, air metros, and air taxis**, see the ³

³The terms refer to different AAM applications: (i) **last-mile delivery** are logistical operations that use unmanned aircraft to deliver cargo from the final distribution center to the consumer; (ii) **air metros** refer to fixed routes of urban transportation using eVTOLs in predetermined air corridors, functioning as an alternative to conventional ground public transportation systems; and (iii) **air taxis** consist of on-demand services that transport passengers between user-selected locations, providing greater route flexibility.

Similarly, Morgan Stanley Research [5] emphasizes that AAM regulation must consider the differences between various types of aircraft and their implications for traffic and airspace management systems.

Complementing this perspective, Ravich [8] argues that on-demand air mobility will require legislators to address additional challenges, including **airspace management, property rights, environmental impacts, privacy, and cybersecurity**.

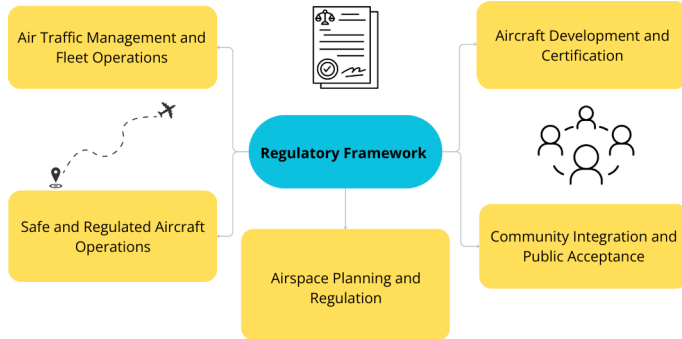


Figure 1.6: Categories of AAM Regulation.

From another perspective, the Australian Unmanned Systems Association (AAUS) [2], in its 2021 report on AAM, identifies six strategic pillars essential for implementing this new paradigm highlighted in Figure 1.7.

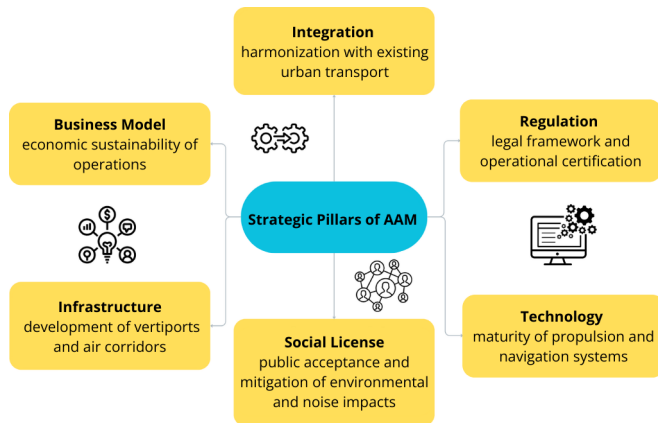


Figure 1.7: Strategic pillars of AAM.

In synthesis, **the evolution of the AAM ecosystem will therefore depend on a holistic and adaptive regulatory approach that balances technological innovation with operational safety.** The maturation of this regulatory framework will be essential to enable the full integration of advanced air mobility into the existing urban landscape, enabling not only new transportation models, but also a profound transformation in the conception of urban mobility for the coming decades. The role of regulatory agencies, in collaboration with industry and academia, will be decisive in establishing an environment conducive to innovation, without compromising airspace safety and the needs of benefiting communities.

1.4 Aircraft in the AAM Ecosystem

To achieve AAM objectives, **new aircraft are being designed** specifically to operate in this differentiated airspace context. Such aircraft possess characteristics distinct from traditional airplanes and helicopters, offering solutions adapted to urban and suburban operations.

An essential category in this ecosystem is eVTOL aircraft (*electric Vertical Take-off and Landing*), that is, electric aircraft with vertical takeoff and landing capability. These are designed specifically for **integration in densely populated urban areas, due to their reduced size, lower noise levels, and reduced emissions.**

Additionally, Unmanned Aircraft Systems (UAS) are also noteworthy. These operate autonomously or with minimal human supervision, significantly expanding operational possibilities, such as in rapid deliveries, infrastructure inspections, and urban aerial monitoring.

Unlike traditional aircraft, AAM aircraft have **reduced size and operate at lower altitudes**, making their tracking or continuous control through conventional technologies, such as radar or satellite, unfeasible. Such a condition demands the **implementation of new systems for monitoring and managing air traffic.**

For a view of terminology, Table 1.2 presents the technical terminology of the main aircraft that compose the AAM ecosystem, providing

simplified definitions that facilitate the understanding of the addressed concepts.

Term	Definition
Conventional takeoff and landing - CTOL	Fixed-wing aircraft that requires conventional runway for takeoff and landing.
Short takeoff and landing - STOL	Aircraft that requires short runways for takeoff and landing operations.
Vertical takeoff and landing - VTOL	Aircraft capable of taking off, hovering, and landing vertically without the need for a runway. VTOL aircraft frequently use electric propulsion (eVTOL).
Rotary-wing aircraft	Aircraft that generates lift through rotating rotors, such as helicopters and autogyros.
Unmanned Aircraft Systems - UAS	Unmanned aircraft and its related components, such as control and communication systems, essential for its safe and efficient operation.

Table 1.2: Terminology of Aircraft in the AAM Ecosystem

The aeronautical configuration most highlighted in the current phase of AAM development is the one called **Lift+Cruise**. A conceptual example is displayed in Figure 1.8.



Figure 1.8: Concept design – NASA Lift+Cruise

Source: Adapted from <https://evtol.news/nasa-lift-cruise>.

The *Lift+Cruise* aircraft operates in two main distinct modes:
Main operational modes.

- **Vertical takeoff and landing:** uses specific propellers for operations in restricted areas.
- **Horizontal cruise flight:** other propellers assume control, providing greater efficiency in horizontal flight.

This configuration leverages **operational flexibility similar to helicopters with the aerodynamic efficiency of traditional fixed-wing aircraft**. Additionally, these aircraft can be operated by human pilots or autonomously, offering operational versatility.

The predominance of characteristics similar to fixed-wing aircraft confers important advantages in the **regulatory certification process**. Because aircraft with known characteristics and widely regulated features possess additional facilities during the **certification process**.

Another example of prominence in AAM development is the **electric helicopter Quiet Single Main Rotor (QSMR)** developed by NASA, illustrated in Figure 1.9. This **eVTOL helicopter** is being designed with special focus on **noise reduction**, capable of carrying up to six passengers and operated either piloted or autonomously.



Figure 1.9: Concept design - NASA Quiet Single Main Rotor (QSMR)

Source: Adapted from <https://evtol.news/nasa-quiet-main-rotor>.

The QSMR utilizes electric or hybrid propulsion and includes technological features to minimize operational noise. Among such features

stand out the **main propellers with special design for acoustic reduction**, similar to the well-known NASA Blue Edge propellers, recognized for generating **less noise and greater aerodynamic efficiency**.

Finally, Figure 1.10 illustrates a complete AAM ecosystem, showing different types of eVTOL aircraft operations in urban and rural environments.



Figure 1.10: Ecosystem of Operations and Aircraft of AAM.

1.5 Integrated Learning Section

Upon completing this section, the student should be able to: critically understand the conceptual fundamentals of AAM/UAM; identify technological, regulatory, and social challenges; relate aeronautical configurations with impacts on vertiports and airspace; develop evidence-based technical arguments grounded in scientific literature; and apply theoretical concepts to operational and regulatory analysis.

Block 1 — Theoretical Foundation and Critical Analysis

Required readings.

1. Garrow, L. A., et al. (2021). *Urban air mobility: A comprehensive review and comparative analysis with autonomous and electric ground transportation for informing future research.*
2. Kieseewetter, P., et al. (2023). *A holistic review of the current state of research on aircraft design concepts and consideration for advanced air mobility applications.*
3. Pak, H., et al. (2024). *Can Urban Air Mobility become reality? Opportunities and challenges of UAM as innovative mode of transport and DLR contribution to ongoing research.*
4. Arafat, M. Y., & Pan, S. (2024). *Urban Air Mobility Communications and Networking: Recent Advances, Techniques, and Challenges.*

Written assignment (up to 2 pages). Develop a structured comparative analysis addressing:

1. Technical convergences among authors;
2. Divergences regarding feasibility, technological maturity, and timeline;
3. Main regulatory bottlenecks identified;
4. Technology considered most critical for AAM viability — provide technical justification.

The analysis must demonstrate synthesis, critical thinking, and conceptual articulation. This is not a summary, but a substantiated positioning.

Block 2 — Operational and Infrastructural Application

Applied technical analysis (up to 2 pages). Discuss how the aeronautical configurations *Lift+Cruise*, *Tilt-wing*, *Tail-sitter*, and conventional rotary-wing impact the following aspects:

1. Physical sizing of vertiports;

2. Operational separations and procedures;
3. Energy requirements and ground times;
4. Complexity of urban airspace management;
5. Integration with ATM/UTM systems.

The analysis must establish causal relationships between aerodynamic characteristics, operational envelope, and required infrastructure.

Block 3 — Regulatory Framework and Planning

Strategic discussion. Answer in a structured manner:

1. How do current regulations (FAA, EASA, ANAC) facilitate or limit AAM?
2. What community risks must be considered in urban planning?
3. How to balance technological innovation and operational safety?
4. Is AAM a solution for rural mobility or does it remain centered in large urban centers?

Block 4 — Structured Technical Exercise

Comparative technical study. Select one real or conceptual example for each category: CTOL, STOL, VTOL (eVTOL), Rotary-Wing, and UAS. For each aircraft, present:

1. Range, speed, and capacity;
2. Type of propulsion;
3. Predominant operational environment;
4. Associated regulatory challenges;
5. Implications for vertiports or aerodromes.

Block 5 — Consolidation Questionnaire

1. List three critical urban challenges for AAM implementation.
2. Operationally differentiate *Sub-Urban-Commuter* and *Airport-Shuttle*.
3. What factors influence public acceptance?
4. Which three infrastructures are indispensable to the AAM ecosystem?
5. Which subsystem (UAM, RAM, or Rural) presents the greatest regulatory complexity? Justify.

2

COURSE SYLLABUS AND PROJECT

2.1 Course Description

The course addresses the fundamentals and applications of Urban Air Mobility (UAM), presenting an integrated view of the technical, regulatory, operational, and planning aspects associated with the implementation of this new transportation system. Initially, the global UAM landscape, the technological evolution of eVTOL aircraft, and the economic structure of the sector are discussed, including the main ecosystem agents and emerging business models.

Subsequently, the institutional and regulatory aspects that structure the development of urban air mobility are analyzed, with emphasis on aircraft certification, airspace regulation, and the competencies of national and international aeronautical authorities.

The course also addresses demand modeling methodologies and spatial analysis techniques applied to transportation system planning, including origin-destination studies, discrete choice models, spatial demand distribution, and identification of trip-generating centers. Tools for territorial analysis and Geographic Information Systems (GIS) are introduced to support the assessment of urban and environmental constraints and site selection for vertiport deployment.

In sequence, the operational concepts of vertiports and the operational characteristics of eVTOL aircraft are studied, including system operational requirements, capacity estimation, and the concept of Trajectory-Based Operations (TBO), used for planning and managing air operations based on predictable trajectories integrated into airspace management. Principles of airside sizing are also discussed, including the definition of FATO, TLOF, and Safety Areas, the planning of approach and departure routes, and the definition of operational protection zones, obstacle limitation surfaces, and noise protection zones.

Finally, the course presents the fundamentals of vertiport architectural design and its integration into the urban context, including passenger terminal sizing, the study of operational flows, levels of service, and multimodal integration with ground transportation systems. Sustainability aspects, environmental and social impacts, and infrastructure implementation and expansion strategies are also discussed.

Throughout the semester, students develop a complete vertiport project, integrating market analysis, demand estimation, site selection, operational planning, and architectural design of the infrastructure.

2.2 Activity Schedule

03/03 – Lecture 1: Urban Air Mobility Landscape

- Global and historical overview of Urban Air Mobility (UAM);
- Fundamental concepts and terminology;
- Technological evolution of eVTOL aircraft;
- International trends and sector perspectives.

10/03 – Lecture 2: UAM Market Structure

- Economic structure of the UAM industry;
- Main ecosystem agents;

- Supply chain and business models;
- Market perspectives and implementation challenges.

17/03 – Lecture 3: Regulatory and Normative Framework

- International regulatory framework (FAA and EASA);
- Brazilian regulatory framework (ANAC and DECEA);
- Normative requirements applicable to vertiports;
- Regulatory impacts on infrastructure implementation.

24/03 – Lecture 4: Demand Modeling for UAM

- Fundamentals of demand modeling in transportation;
- Discrete choice models (Logit);
- Stated and revealed preference models;
- Origin-destination studies;
- Time-price elasticity;
- Demand scenarios for UAM systems.

31/03 – Lecture 5: Spatial Structure of Demand

- Spatial distribution of urban demand;
- Gravity models;
- Identification of trip-generating centers;
- Definition of system coverage radius;
- Introduction to territorial analysis applied to vertiport planning.

07/04 – Lecture 6: Introduction to GIS for Territorial Analysis and Environmental Impacts

- Fundamentals of Geographic Information Systems (GIS);
- Geospatial databases;
- Spatial analysis applied to urban air mobility;
- Identification of environmental and urban constraints;
- GIS applications in infrastructure planning.

14/04 – Lecture 7: Vertiport Site Selection

Guest lecture: Msc. Marcelo Saraiva Peres (PhD Candidate – ITA)

- Technical criteria for site selection;
- Accessibility and multimodal connectivity;
- Multi-criteria analysis methods (AHP / PROMETHEE);
- GIS applications for territorial evaluation.

25/04 – Partial Delivery 1 Presentation

Presentation of the **Feasibility Report** containing:

- evidence-based demand analysis;
- site selection with technical justification;
- application of technical and operational feasibility criteria;
- discussion of analyzed alternatives and study limitations.

Each group will present preliminary project results and will receive technical feedback for the development of subsequent course phases.

28/04 – Recovery Week

No lecture.

05/05 – Lecture 8: Introduction to Vertiport Operations

- Operational concept of vertiports;

- Types of eVTOL aircraft and their specifications;
- Operational parameters (range, speed, payload);
- Definition of operational requirements based on estimated demand.

12/05 – Lecture 9: Operational Capacity and Trajectory-Based Operations

- Concept of **TBO** (***Trajectory-Based Operations***);
- Planning and management of trajectory-based operations;
- Sequencing and predictability of operations;
- Relationship between TBO, operational capacity, and system performance;
- Integration between vertiport operations and airspace management.

19/05 – Lecture 10: Airside Sizing

- Definition and positioning of **FATO**, **TLOF**, and **Safety Area**;
- Operational configuration of the vertiport;
- Planning of approach and departure routes;
- Preliminary obstacle analysis.

26/05 – Lecture 11: Operational Protection Zones

- Design of vertiport protection zones;
- Obstacle limitation surfaces;
- Noise protection zones;
- Operational impacts in the urban environment.

02/06 – Lecture 12: Support Infrastructure

- Energy systems and eVTOL charging infrastructure;
- Maintenance infrastructure;
- Operational safety;
- Emergency procedures;
- Environmental requirements.

Partial Delivery 2

Technical design of the airside containing:

- vertiport layout;
- operations plan;
- preliminary simulations.

09/06 – Lecture 13: Introduction to Architectural Design

- Fundamentals of vertiport architectural design;
- Passenger terminal sizing;
- Integration with urban transportation;
- Multimodal integration.

16/06 – Lecture 14: Passenger Flows and Terminal Operations

- Passenger processing;
- Terminal operational flows;
- Processing times;
- Levels of service;
- Terminal operational simulation.

23/06 – Lecture 15: Project Consolidation

- Review and refinement of the complete project;
- Implementation planning;
- Future expansion strategies;
- Sustainability and urban integration.

30/06 – Final Project Presentation

Final presentation of the complete vertiport project, including:

- market analysis;
- demand estimation;
- site selection;
- operational design;
- architectural design.

2.3 Project

The course development will be organized in the format of a Digital Engineering Studio. In this pedagogical model, students develop an applied project throughout the semester, continuously documenting their analyses and engineering decisions in a structured digital environment.

Each group will develop a conceptual project for implementing a vertiport in a selected city as a case study. The objective of the project is to apply, in an integrated manner, the concepts discussed in class related to urban air mobility, including demand analysis, territorial planning, operational definition, and architectural design of the infrastructure.

The project will be developed progressively throughout the semester, following the logical sequence of infrastructure planning: feasibility analysis, definition of the operational concept, and development of the integrated vertiport design.

The work will be carried out in groups and structured in two partial deliveries and a final project.

Partial Delivery 1 – Feasibility Analysis

In this stage, students will develop a preliminary feasibility study for vertiport implementation.

The work should present:

- characterization of the selected city for the study;
- preliminary analysis of the market and the potential for Urban Air Mobility utilization;
- initial demand estimation, based on population data, mobility characteristics, or urban features;
- identification of possible areas for vertiport implementation;
- selection of a preferred site, accompanied by technical justification.

The objective of this stage is to demonstrate that the chosen location presents suitable initial conditions for implementation of the proposed infrastructure.

Partial Delivery 2 – Technical Planning of the Vertiport

The second stage consists of developing the technical and operational concept of the selected vertiport.

The work should present:

- description of the vertiport's operational concept;
- estimation of the number of operations expected based on estimated demand;
- preliminary proposal for vertiport layout;

- identification of landing and takeoff areas (FATO, TLOF, and safety areas);
- description of approach and departure routes.

The objective of this stage is to demonstrate the initial technical feasibility of the proposed infrastructure.

Final Project

In the final stage, students will consolidate the project developed throughout the semester, integrating the results obtained in the previous stages.

The final project should present:

- synthesis of market analysis and demand estimation;
- justification for site selection;
- technical layout of the vertiport;
- architectural proposal for the passenger terminal;
- description of the system's operational functioning;
- integration of the vertiport with the urban context and with other transportation modes.

Project Portal

Each group will develop a digital portal to document the project throughout the semester. This portal will function as a dynamic technical report, in which the results of analyses, maps, diagrams, layouts, and other elements produced during the course are progressively recorded.

In this way, the project will be structured as a living report, continuously updated as activities advance. At the end of the semester, navigation through the project portal should allow complete understanding of the developed project, from the initial feasibility analysis to the vertiport's final proposal.

The final presentation will consist of guided navigation through the project portal, enabling integrated visualization of the analyses conducted, the planning decisions adopted, and the final solution proposed by the group.

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